Swati Aggrawal

swatiagg357@gmail.com | +39 3338004120 | Rome,Italy | GitHub | LinkedIn

EDUCATION

Sapienza Università di Roma **Expected: May 2026**
Master's in Computer Science *Rome, Lazio*

- **Coursework:** Deep Learning And Applied AI, NLP, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning (Ongoing)
- Selected for a competitive international master's program

Sharda University **July 2023**
Bachelor of Technology - CSE with AI and ML *Greater Noida*

- **Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity To Computers In Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort)

RESEARCH EXPERIENCE

Machine Learning for Air Pollution — May 2023

- Published a **peer-reviewed IEEE ViTECoN-2023 paper** evaluating ML models (RF, SVM, GA, LS-SVM, ANN) for air quality prediction, demonstrating improved performance over traditional statistical methods.

Continual Learning for Computer Vision — May 2025

- Conducted a structured literature review of **60+ peer-reviewed papers** on continual learning in computer vision, covering catastrophic forgetting, replay-based and regularization approaches, and TIL/DIL/CIL protocols.

PROJECTS

Lossy Text Compressor

- Implemented deep learning based lossy text compression using **semantic embeddings (GTR-T5 Base)** and multi-level quantization
- Evaluated reconstruction quality via **BLEU, ROUGE-L, and cosine similarity**, showing **6–8 bit quantization** achieves best accuracy–efficiency balance

ChatBot Using NLTK

- Designed and implemented a rule-based NLP chatbot using Python and NLTK, leveraging **tokenization, lemmatization, POS tagging,** and pattern-matching algorithms for intent recognition and response generation.

PosePal

- Developed and deployed a deep learning based yoga pose classification system using **CNNs and transfer learning** with TensorFlow/Keras, integrating the trained model into a **web-based application** for real-time pose recognition.

SKILLS

- **Technical:** Python, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, CNNs, Transformers, TensorFlow, Keras, Git, GitHub
- **Core Strengths:** Analytical problem solving, research communication, independent learning